TANGENT SPACES BASE-CHANGE ALONG FLAT RING MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1698-1705 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1707-1793, the complete author proof (85 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_stacks-stacks-more-morphisms.json (pb=3601, stmt 1698-1705, proof 1707-1793).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DNN: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06D3: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-sheaf-of-infinitesimal-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1607–1618. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087R: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05ZF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0617: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Q6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06XH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1793–1793. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack over a scheme S whose structure morphism $\mathcal{X} \to S$ is locally of finite presentation. Let $A \to B$ be a flat S-algebra homomorphism. Let x be an object of $\mathcal{X}$ over A. Then $T_x(M) \otimes_A B = T_y(M \otimes_A B)$.*

Proof. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. We first reduce the lemma to the case where x lifts to U. Recall that $T_x(M)$ is the set of isomorphism classes of lifts of x to $A[M]$. Therefore Lemma lemma-sheaf-of-infinitesimal-lifts[1] says that the rule

$$A_1 \mapsto T_{x|_{A_1}}(M \otimes_A A_1)$$

is a sheaf on the small étale site of $\mathrm{Spec}(A)$; the tensor product is needed to make $A[M] \to A_1[M \otimes_A A_1]$ a flat ring map. We may choose a faithfully flat étale ring map $A \to A_1$ such that $x|_{A_1}$ lifts to a morphism $u_1 : \mathrm{Spec}(A_1) \to U$, see for example Sheaves on Stacks, Lemma 06XH. Write $A_2 = A_1 \otimes_A A_1$ and set $B_1 = B \otimes_A A_1$ and $B_2 = B \otimes_A A_2$. Consider the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & T_y(M \otimes_A B) & \longrightarrow & T_{y|_{B_1}}(M \otimes_A B_1) & \longrightarrow & T_{y|_{B_2}}(M \otimes_A B_2) \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & T_x(M) & \longrightarrow & T_{x|_{A_1}}(M \otimes_A A_1) & \longrightarrow & T_{x|_{A_2}}(M \otimes_A A_2) \end{array}$$

The rows are exact by the sheaf condition. We have $M \otimes_A B_i = (M \otimes_A A_i) \otimes_{A_i} B_i$. Thus if we prove the result for the middle and right vertical arrow, then the result follows. This reduces us to the case discussed in the next paragraph.

Assume that x is the image of a morphism $u : \mathrm{Spec}(A) \to U$. Observe that $T_u(M) \to T_x(M)$ is surjective since $U \to \mathcal{X}$ is smooth and representable by algebraic spaces, see Criteria for Representability, Lemma 06D3 (see discussion preceding it for explanation) and More on Morphisms of Spaces, Lemma 04AM. Set

[1] This lemma applies: $\Delta : \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is locally of finite presentation by Morphisms of Stacks, Lemma 06Q6 and the assumption that $\mathcal{X} \to S$ is locally of finite presentation. Therefore $\mathcal{I}_{\mathcal{X}} \to \mathcal{X}$ is locally of finite presentation as a base change of Δ.

$R = U \times_{\mathcal{X}} U$. Recall that we obtain a groupoid (U, R, s, t, c, e, i) in algebraic spaces with $\mathcal{X} = [U/R]$. By Artin's Axioms, Lemma 0DNN we have an exact sequence

$$T_{e \circ u}(M) \to T_u(M) \oplus T_u(M) \to T_x(M) \to 0$$

where the zero on the right was shown above. A similar sequence holds for the base change to B. Thus the result we want follows if we can prove the result of the lemma for $T_u(M)$ and $T_{e \circ u}(M)$. This reduces us to the case discussed in the next paragraph.

Assume that $\mathcal{X} = X$ is an algebraic space locally of finite presentation over S. Then we have

$$T_x(M) = \mathrm{Hom}_A(x^*\Omega_{X/S}, M)$$

by the discussion in More on Morphisms of Spaces, Section 0617. By the same token

$$T_y(M \otimes_A B) = \mathrm{Hom}_B(y^*\Omega_{X/S}, M \otimes_A B)$$

Since $X \to S$ is locally of finite presentation, we see that $\Omega_{X/S}$ is locally of finite presentation, see More on Morphisms of Spaces, Lemma 05ZF. Hence $x^*\Omega_{X/S}$ is a finitely presented A-module. Clearly, we have $y^*\Omega_{X/S} = x^*\Omega_{X/S} \otimes_A B$. we conclude by More on Algebra, Lemma 087R. □